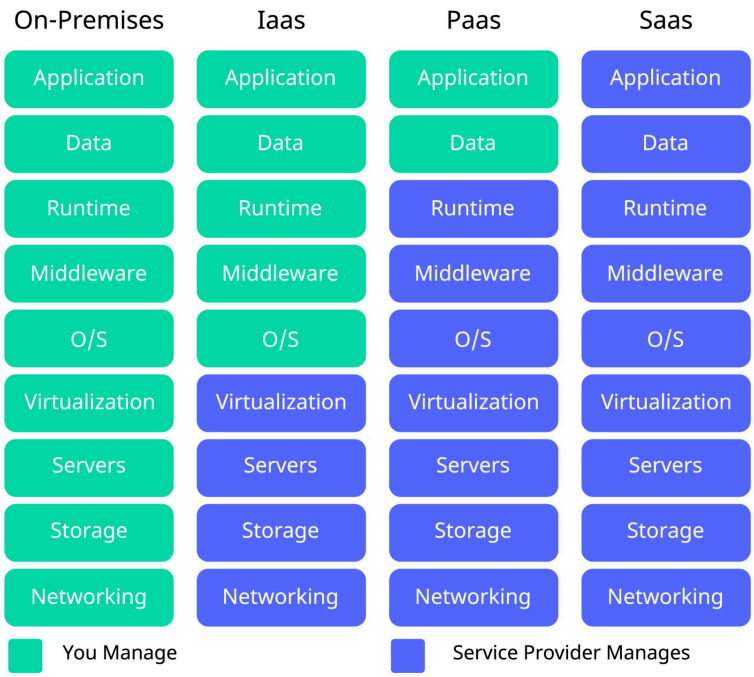
Core Cloud Service Models

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User prompt: Cloud Service Models

Response: Cloud service models categorize the level of control, management, and flexibility you have when utilizing cloud computing resources. They define the **Shared Responsibility Model**—how duties are divided between you (the customer) and the cloud service provider.



Core Cloud Service Models

1. Infrastructure as a Service (IaaS)

IaaS provides basic computing infrastructure—virtual servers, raw storage, and networking components—over the internet. You get maximum control over your IT environment without buying or maintaining physical hardware.

- **You Manage:** Operating Systems (OS), runtime, middleware, data, and applications.
- **Provider Manages:** Physical data centers, virtualization layer, servers, networking hardware, and storage drives.
- **Best For:** Migrating existing legacy workloads (lift-and-shift), network architecture customization, and high-performance computing needs.
- **Examples:** AWS (EC2), Microsoft Azure (VMs), Google Cloud Compute Engine.

2. Platform as a Service (PaaS)

PaaS offers a complete development and deployment environment in the cloud. The provider manages the underlying infrastructure and operating system, leaving developers free to focus

solely on writing and deploying code.

- **You Manage:** Application code, business logic, and application-level data.
- **Provider Manages:** Hardware, virtualization, OS, runtime environments, middleware, and database engines.
- **Best For:** Rapid application development, API creation, and microservices architectures without DevOps overhead.
- **Examples:** Heroku, AWS Elastic Beanstalk, Google App Engine.

3. Software as a Service (SaaS)

SaaS delivers fully functional, end-user applications over the web on a subscription or pay-per-use basis. You interact directly with the software interface through a browser or lightweight app.

- **You Manage:** User access management, permissions, and initial configuration settings.
- **Provider Manages:** Everything—infrastructure, operating system, application updates, security patches, and performance scaling.
- **Best For:** Everyday productivity, collaboration, customer relationship management (CRM), and standard enterprise software.
- **Examples:** Google Workspace, Microsoft 365, Salesforce, Dropbox.

Comparative Breakdown

Service Model	Management Control	Target Audience	Primary Focus	Key Benefit
IaaS	High	System Administrators & DevOps	Virtualized Infrastructure	Maximum customization & flexibility
PaaS	Moderate	Developers & Software Engineers	Code & Application Logic	Speeds up development cycle
SaaS	Low	End Users & Business Operations	Ready-to-use Software	Zero maintenance & quick rollout

Modern Emerging Service Models

Beyond the primary three, specialized models have emerged to optimize workflows:

- **Function as a Service (FaaS) / Serverless:** Event-driven execution where code runs only when triggered by specific events (e.g., AWS Lambda, Google Cloud Functions). Providers bill strictly for execution time down to the millisecond.
- **Backend as a Service (BaaS):** Provides cloud-backed services for mobile and web apps (like authentication, database sync, and push notifications) so front-end developers don't

have to build server logic (e.g., Firebase).

- **Artificial Intelligence as a Service (AIaaS):** Delivers pre-trained AI and machine learning models via APIs (vision recognition, language models, sentiment analysis) without needing in-house data science teams.

Sources

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5. What Are Cloud Services? Types, Deployment Models &... - SUSE
